Leo Tsukada

Email: leo.tsukada@ligo.org
Website: leotsukada.github.io
LinkedIn: leo-tsukada
Work authorization: H-1B;
self-petitioned EB-2 pending,
EAD expected 2027

Employment

University of Wisconsin–Milwaukee — Milwaukee, WI
Postdoctoral Research Associate — 2026–present

University of Nevada, Las Vegas — Las Vegas, NV
NCfA Postdoctoral Fellow — 2024–2026

Pennsylvania State University — University Park, PA
Assistant Research Professor — 2022–2024
Postdoctoral Scholar — 2021–2022

Education

The University of Tokyo — Tokyo, Japan
Ph.D. in Physics, Advisor: Prof. Kipp Cannon — 2018–2021

- Thesis title : Modeling and Searching for Stochastic Gravitational-waves Backgrounds from Ultralight Boson Particles

The University of Tokyo — Tokyo, Japan
M.S. in Physics, Advisor: Prof. Kipp Cannon — 2016–2018

- Thesis title : Towards a Search for Stochastic Gravitational-Wave Backgrounds from Ultra-light Bosons

The University of Tokyo — Tokyo, Japan
B.S. in Applied Physics, Advisor: Prof. Norikatsu Mio — 2011–2016

- Thesis title : Performance evaluation of the frequency reference cavity for KAGRA detector

Research Visits

LIGO Lab, California Institute of Technology — Pasadena, CA
LIGO visitor program, hosted by Prof. Alan Weinstein — Summer 2019

Laboratoire d'Annecy de Physique des Particules — Annecy, France
Visiting research, supervised by Dr. Tania Regimbau — Fall 2018

University of Minnesota — Minneapolis, MN
Visiting research, supervised by Prof. Vuk Mandic — Spring 2018

LIGO Livingston Observatory, California Institute of Technology — Livingston, LA
LIGO SURF program, supervised by Dr. Valery Frolov — Summer 2014

Research Experience

Low-latency gravitational wave (GW) search pipeline: GstLAL, SGNL — 2016–present
UTokyo, Caltech, Penn State, UNLV, UWM

- Fast evaluation of trigger consistency between multiple detectors
- Operation and event follow-up for online analysis
- Optimization of the template-bank compression

- Improvement of statistical post-processing
- GPU acceleration of the GW search pipeline

Targeted GW search for electromagnetic counterparts 2024–present
UNLV, UWM

- Optimization of the GstLAL search targeted for electromagnetic counterparts
- Deployment of the targeted GW search pipeline into the GW observing run

Compact-binary population analysis with GW events 2024–present
UNLV, UWM

- Incorporation of sub-threshold events into the population analysis

Anisotropic stochastic gravitational wave background (SGWB) search 2017–present
LAPP, Penn State, UNLV, UWM

- Mock data study for anisotropic background from compact binary coalescences
- Development and operation of the LVK search for anisotropic SGWB
- Bayesian parameter estimation for targeted anisotropic SGWB
- Extension of the anisotropic SGWB search to include non-GR polarization, time dependency

Searching for ultra-light bosons through SGWB 2016–2021
UTokyo, UMN, Caltech

- Modeling SGWB spectrum from ultra-light scalar/vector field through BH superradiance
- Developing a search pipeline for SGWB from ultra-light scalar fields

Early research: detector instrumentation 2014–2016
LIGO Livingston, UTokyo

- Theoretical model of the optical loss inside the Advanced LIGO arm cavities
- Optical properties and frequency stability of the KAGRA reference cavity

Teaching Experience

- **Substitute Lecturer** at Pennsylvania State University Fall 2022
  Electromagnetism
- **Teaching Assistant** at The University of Tokyo Fall 2016
  Analytical mechanics

Fellowship and Awards

- **University Faculty Travel Award**, University of Nevada Las Vegas 2025
- **NCfA Fellowship**, University of Nevada Las Vegas 2024
- **Paper Award**, United Japanese researchers Around the world 2024
- **LEADER Fellowship** (declined), Japan Society for the Promotion of Science (JSPS) 2023
- **Best Poster Award**, Gravitational Wave Orchestra 2022
- **Best Presentation Award**, The 7th KAGRA International Workshop 2020
- **DC1 Research Fellowship**, JSPS 2018–2021
- **LIGO Visitor Program**, California Institute of Technology 2019
- **Overseas Challenge Program for Young Researchers**, JSPS 2019
- **GRASP Scholarship**, The University of Tokyo 2018
- **SURF Program**, California Institute of Technology 2014
- **Best Project Award**, Cosmic/Particle Spring school 2014

Research Talks

Invited talks, seminars, panel, symposium

Expanding the Potential of Gravitational-Wave Astronomy From Detections to Demographics and Future Directions — 2026
TEPAPP seminar, 2026, UCLA — Los Angeles, USA

Panelist for Gravitational-wave physics session — 2026
NCfA Symposium 2026, UNLV — Las Vegas, USA

Panel chair for multi-messenger and multiband astrophysics — 2025
PAX+CE symposium 2025, UIUC — Urbana-Champaign, USA

Panelist for Astrophysics of Compact Binary Mergers — 2025
NCfA Symposium 2025, UNLV — Las Vegas, USA

Going wider and deeper in the search for gravitational waves — 2024
CRA seminar, Georgia Institute of Technology — Atlanta, USA

Going wider and deeper in the search for gravitational waves — 2024
Astrophysics Colloquium, Texas Tech University — Lubbock, USA

Overview and prospect of the GW transient search in the fourth observing run — 2024
The extreme Universe : CTA-Japan workshop — Tokyo, Japan

Toward unified Bayesian parameter inference of stochastic gravitational wave backgrounds — 2023
LIGO seminar, California Institute of Technology — Pasadena, USA

The improvement of GstLAL's ranking statistics toward the fourth observing run — 2023
Utrecht & UMass Dartmouth joint seminar — USA (online)

Overview and future prospect of LIGO-Virgo-KAGRA's fourth observing run — 2023
Astronomy Society of Japan Autumn meeting — Nagoya, Japan

Toward unified Bayesian parameter inference of stochastic gravitational wave backgrounds — 2023
C-lab seminar, Nagoya University — Nagoya, Japan

Panelist for gravitational waves and multi-messenger astronomy — 2023
New Evolution of Multi-Messenger Astrophysics, Penn State — State College, USA

Observation of neutron stars during LIGO-Virgo-KAGRA's observing runs — 2022
APS April meeting — New York, USA

Modeling and searching for a stochastic GW background from ultralight bosons — 2021
GW Physics and Astronomy: Genesis, The Fourth Annual Area Symposium — Japan (online)

Low-latency detection of the GWs from compact binary coalescences — 2022
ISAS seminar, ISM astronomy seminar, JGW seminar — Japan

Gravitational waves from neutron star-black hole coalescences — 2021
LIGO-Virgo-KAGRA Collaboration webinar — online

First observations of black hole and neutron star mergers — 2021
Fundamental Theory Seminar, Penn State — Pennsylvania, USA

First search for stochastic GW backgrounds from ultra-light bosons — 2018
The CGCA seminar, University of Wisconsin Milwaukee — Wisconsin, USA

Application of a low-latency whitening filter to CBC GW searches — 2016
RESCEU joint seminar, The University of Tokyo — Tokyo, Japan

Selected contributed talks

Targeted gravitational-wave search towards medium-latency followup. 2026
APS April meeting Denver, USA

Targeted gravitational-wave search for compact binaries using a GstLAL pipeline 2025
NCfA Symposium 2025, UNLV Las Vegas, USA

The improvement of GstLAL's ranking statistics toward the fourth observing run 2023
APS April meeting Minnesota, USA

First observations of black hole and neutron star mergers 2021
The 8th KAGRA International Workshop Korea (online)

Modeling and searching for a stochastic GW background from ultralight bosons 2021
Amaldi 14 Australia (online)

Stochastic GW backgrounds from ultra-light vectors 2019
The 29th Workshop on General Relativity and Gravitation in Japan Kobe, Japan

Anisotropic GW background Mock data study 2019
Gravitational Wave Physics and Astronomy Workshop Tokyo, Japan

A first search for stochastic GW backgrounds from ultra-light scalars 2018
Gravitational Wave Physics and Astronomy Workshop Maryland, USA

Application of a low-latency whitening filter to CBC GW searches 2017
The Third KAGRA International Workshop Taipei, Taiwan

Leadership / Professional Service

- **Editorial chair of gravitational wave transient catalog (GWTC)-5 Results paper** : LVK Collaboration 2025–present
- **Organizer of GstLAL face-to-face workshop** : University of Nevada, Las Vegas 2025
- **Organizer of KAGRA GW seminar** 2024–present
- **Co-chair of GWTC working group** : LVK Collaboration 2024
- **Co-chair of anisotropic stochastic-background working group** : LVK Collaboration 2022–present
- **Thesis committee** : Pennsylvania State University 2023
- **Referee** : Physical Review D, Physical Review Letter 2022–present
- **Advanced LIGO science summaries** : Writer and japanese translator 2021
- **Vice director** : Cosmic/Astrophysics Student Summer School in Japan 2019
- **Workshop Assistant** : Gravitational Wave Physics and Astronomy Workshop 2019
- **LVK Collaboration services** : See https://laac.docs.ligo.org/recognition/2024/09/08/leo-tsukada 2017–present

Mentoring Experience

- **Maxwell Siebersma** Ph.D student at Luisian State University summer 2026 - present
  Fast targeted gravitational-wave search for compact binaries
- **Aditya Ramdasi** Ph.D student at University of Wisconsin-Milwaukee spring 2026 - present
  Fast targeted gravitational-wave search for compact binaries
- **Pratyusava Baral** Ph.D student at University of Wisconsin-Milwaukee spring 2026
  Production of gravitational-wave transient catalogs for GstLAL detection pipeline
- **Izumi Kaku** Ph.D student at Osaka Metropolitan University spring 2025

Searches for anisotropic gravitational-wave background with circular polarizations

- **Jonathan Regan** Ph.D student at University of Nevada Las Vegas — fall 2024 - present
  Incorporation of sub-threshold events into the CBC population analysis
- **Noah Zhang** Ph.D student at Georgia Institute of Technology — spring 2024 - present
  Fast targeted gravitational-wave search for compact binaries
- **Soichiro Kuwahara** Ph.D student at The University of Tokyo — spring 2022 - fall 2024
  Component separation analysis for anisotropic gravitational-wave backgrounds
- **Santiago Jaraba** Ph.D student at Universidad Aut'onoma de Madrid — spring 2022 - spring 2024
  Parameter estimation for anisotropic gravitational-wave backgrounds [18]
- **Deepali Agarwal** Ph.D student at IUCAA — spring 2022 - 2023
  Parameter estimation for anisotropic gravitational-wave backgrounds [18]
- **Erik Floden** Ph.D student at University of Minnesota — spring 2021 - fall 2024
  Parameter estimation and spherical-harmonics searches of anisotropic gravitational-wave backgrounds [18], [21], [37]
- **Anarya Ray** Ph.D student at University of Wisconsin-Milwaukee — spring 2022 - spring 2023
  Improving background sampling procedure for GstLAL
- **Richard George** Ph.D student at The University of Texas at Austin — spring 2022 - spring 2023
  Improving SNR $- \xi^2$ *signal model of GstLAL [15]*
- **Andre Guimaraes** Ph.D student at Louisiana State University — spring 2022 - spring 2023
  Improving SNR $- \xi^2$ *signal model of GstLAL [15]*
- **Shio Sakon** Ph.D student at Pennsylvania State University — spring 2022
  Optimization of GstLAL's template bank [10]
- **Shomik Adhicary** Ph.D student at Pennsylvania State University — spring 2022 - spring 2023
  Improving ranking statistics for gravitational-wave detection pipeline, GstLAL [15]
- **Prathamesh Joshi** Ph.D student at Pennsylvania State University — spring 2022 - fall 2024
  Implementation of contamination removal and bank-ξ^2 statistics in GstLAL [14], [15]
- **Takuya Tsutsui** Ph.D student at The University of Tokyo — 2019
  Rapid localization of gravitational wave sources [25]

Outreach

- **"*Out of This World*" Back-to-School event at Las Vegas** — 2025
- **GW Summer Camp at Penn State** — 2024
- **KAGRA outreach group** — 2020–2021
- **SCJSF&JABA forum talk** — 2020
- **Japanese translation of GW190425's science summary** — 2019
- **GW education at a public school in Pasadena** — 2019
- **RESCEU Open Lab** — 2017, 2018
- **International Space Education Board Student Program** — 2015, 2016

Publication List

Short Author List Publications and Preprints

[1] **L. Tsukada**, N. Zhang, S. Sachdev, S. Adhicary, C. Hanna, P. Joshi, and D. Singh, *Fast targeted gravitational-wave followup search for compact binary mergers using gstlal pipeline*, Jun. 2026. arXiv: 2606.22266 [astro-ph.HE].

[2] Y.-J. Huang, C. Hanna, **L. Tsukada**, A. Baylor, O. Godwin, P. Joshi, J. Kennington, C. Messick, S. Sachdev, R. Tapia, and Z. Yarbrough, *Sgnl: Scalable low-latency gravitational wave detection pipeline for compact binary mergers*, 2026. arXiv: 2511.04730 [astro-ph.IM].

[3] Y.-J. Huang, C. Hanna, B. Ewing, P. Godwin, J. Gonsalves, R. Magee, C. Messick, **L. Tsukada**, Z. Yarbrough, *et al.*, "Scalable matched-filtering pipeline for gravitational-wave searches of compact binary mergers", *Phys. Rev. D*, vol. 112, p. 082 002, 8 Oct. 2025.

[4] W. Niu, *et al.* including **L. Tsukada**, *Gw231109_235456: A sub-threshold binary neutron star merger in the ligo-virgo-kagra o4a observing run?*, Sep. 2025. arXiv: 2509.09741 [astro-ph.HE].

[5] P. Joshi, **L. Tsukada**, C. Hanna, *et al.*, *New methods for offline gstlal analyses*, Jun. 2025. arXiv: 2506.06497 [gr-qc].

[6] P. Joshi, W. Niu, C. Hanna, R. Huxford, D. Singh, **L. Tsukada**, *et al.*, *How many times should we matched filter gravitational wave data? a comparison of gstlal's online and offline performance*, May 2025. arXiv: 2505.23959 [gr-qc].

[7] S. Schmidt, S. Caudill, J. D. E. Creighton, R. Magee, **L. Tsukada**, *et al.*, "Searching for gravitational-wave signals from precessing black hole binaries with the gstlal pipeline", *Phys. Rev. D*, vol. 110, p. 023 038, 2 Jul. 2024.

[8] S. S. Chaudhary, *et al.* including **L. Tsukada**, "Low-latency gravitational wave alert products and their performance at the time of the fourth ligo-virgo-kagra observing run", *Proceedings of the National Academy of Sciences*, vol. 121, no. 18, e2316474121, Apr. 2024.

[9] B. Ewing, R. Huxford, D. Singh, **L. Tsukada**, *et al.*, "Performance of the low-latency gstlal inspiral search towards ligo, virgo, and kagra's fourth observing run", *Physical Review D*, vol. 109, no. 4, Feb. 2024.

[10] S. Sakon, **L. Tsukada**, *et al.*, "Template bank for compact binary mergers in the fourth observing run of advanced ligo, advanced virgo, and kagra", *Physical Review D*, vol. 109, no. 4, Feb. 2024.

[11] S. Morisaki, R. Smith, **L. Tsukada**, S. Sachdev, S. Stevenson, C. Talbot, and A. Zimmerman, "Rapid localization and inference on compact binary coalescences with the advanced ligo-virgo-kagra gravitational-wave detector network", *Phys. Rev. D*, vol. 108, p. 123 040, 12 Dec. 2023.

[12] **L. Tsukada**, "Extension of the bayesian searches for anisotropic stochastic gravitational-wave background with nontensorial polarizations", *Physical Review D*, vol. 108, no. 12, Dec. 2023.

[13] S. Banagiri, C. P. L. Berry, G. S. C. Davies, **L. Tsukada**, and Z. Doctor, "Unified p_{astro} for gravitational waves: Consistently combining information from multiple search pipelines", *Phys. Rev. D*, vol. 108, p. 083 043, 8 Oct. 2023.

[14] P. Joshi, **L. Tsukada**, and C. Hanna, "Method for removing signal contamination during significance estimation of a gstlal analysis", *Phys. Rev. D*, vol. 108, p. 084 032, 8 Oct. 2023.

[15] **L. Tsukada**, P. Joshi, *et al.*, "Improved ranking statistics of the gstlal inspiral search for compact binary coalescences", *Physical Review D*, vol. 108, no. 4, Aug. 2023.

[16] A. Renzini, *et al.* including **L. Tsukada**, "Pygwb: A python-based library for gravitational-wave background searches", *The Astrophysical Journal*, vol. 952, no. 1, p. 25, Jul. 2023.

[17] A. Ray, *et al.* including **L. Tsukada**, *When to point your telescopes: Gravitational wave trigger classification for real-time multi-messenger followup observations*, Jun. 2023. arXiv: 2306.07190 [gr-qc].

[18] **L. Tsukada**, S. Jaraba, D. Agarwal, and E. Floden, "Bayesian parameter estimation for targeted anisotropic gravitational-wave background", *Physical Review D*, vol. 107, no. 2, Jan. 2023.

[19] C. Hanna, *et al.* including **L. Tsukada**, "Binary tree approach to template placement for searches for gravitational waves from compact binary mergers", *Physical Review D*, vol. 108, no. 4, 2023.

[20] C. Hanna, *et al.* including **L. Tsukada**, "Metric assisted stochastic sampling search for gravitational waves from binary black hole mergers", *Physical Review D*, vol. 106, no. 8, Oct. 2022.

[21] E. Floden, V. Mandic, A. Matas, and **L. Tsukada**, "Angular resolution of the search for anisotropic stochastic gravitational-wave background with terrestrial gravitational-wave detectors", *Physical Review D*, vol. 106, no. 2, Jul. 2022.

[22] K. Cannon, *et al.* including **L. Tsukada**, "Gstlal: A software framework for gravitational wave discovery", *SoftwareX*, vol. 14, p. 100 680, Jun. 2021, ISSN: 2352-7110.

[23] D. Mukherjee, *et al.* including **L. Tsukada**, "Template bank for spinning compact binary mergers in the second observation run of advanced ligo and the first observation run of advanced virgo", *Physical Review D*, vol. 103, no. 8, Apr. 2021.

[24] **L. Tsukada**, R. Brito, W. E. East, and N. Siemonsen, "Modeling and searching for a stochastic gravitational-wave background from ultralight vector bosons", *Phys. Rev. D*, vol. 103, p. 083 005, 8 Apr. 2021.

[25] T. Tsutsui, K. Cannon, and **L. Tsukada**, "High speed source localization in searches for gravitational waves from compact object collisions", *Phys. Rev. D*, vol. 103, p. 043 011, 4 Feb. 2021.

[26] S. Sachdev, *et al.* including **L. Tsukada**, "An early-warning system for electromagnetic follow-up of gravitational-wave events", *The Astrophysical Journal*, vol. 905, no. 2, p. L25, Dec. 2020.

[27] C. Messick, *et al.* including **L. Tsukada**, *Automating the inclusion of subthreshold signal-to-noise ratios for rapid gravitational-wave localization*, Nov. 2020. arXiv: 2011.02457 [astro-ph.IM].

[28] P. Godwin, *et al.* including **L. Tsukada**, *Incorporation of statistical data quality information into the gstlal search analysis*, Oct. 2020. arXiv: 2010.15282 [gr-qc].

[29] C. Chan, *et al.* including **L. Tsukada**, "Improving the background estimation technique in the gstlal inspiral pipeline with the time-reversed template bank", Sep. 2020. eprint: 2009.03025.

[30] C. Hanna, S. Caudill, C. Messick, A. Reza, S. Sachdev, **L. Tsukada**, *et al.*, "Fast evaluation of multidetector consistency for real-time gravitational wave searches", *Physical Review D*, vol. 101, no. 2, Jan. 2020.

[31] **L. Tsukada**, T. Callister, A. Matas, and P. Meyers, "First search for a stochastic gravitational-wave background from ultralight bosons", *Physical Review D*, vol. 99, no. 10, May 2019.

[32] S. Sachdev, *et al.* including **L. Tsukada**, *The gstlal search analysis methods for compact binary mergers in advanced ligo's second and advanced virgo's first observing runs*, Jan. 2019. arXiv: 1901.08580 [gr-qc].

[33] **L. Tsukada**, K. Cannon, C. Hanna, D. Keppel, D. Meacher, and C. Messick, "Application of a zero-latency whitening filter to compact binary coalescence gravitational-wave searches", *Physical Review D*, vol. 97, no. 10, May 2018.

Collaboration Publications (major contribution)

[34] A. G. Abac , *et al.* including **L. Tsukada**, *Gwtc-5.0: Observations from the second part of the fourth ligo-virgo-kagra observing run and updates to the gravitational-wave transient catalog*, May 2026. arXiv: 2605.27225 [gr-qc].

[35] A. G. Abac , *et al.* including **L. Tsukada**, *Directional search for persistent gravitational waves: Results from the first part of ligo-virgo-kagra's fourth observing run*, 2025. arXiv: 2510.17487 [gr-qc].

[36] A. G. Abac , *et al.* including **L. Tsukada**, *Gwtc-4.0: Updating the gravitational-wave transient catalog with observations from the first part of the fourth ligo-virgo-kagra observing run*, 2025. arXiv: 2508.18082 [gr-qc].

[37] B. P. Abbott, *et al.* including **L. Tsukada**, "Search for anisotropic gravitational-wave backgrounds using data from advanced ligo and advanced virgo's first three observing runs", *Phys. Rev. D*, vol. 104, p. 022 005, 2 Jul. 2021.

[38] B. P. Abbott, *et al.* including **L. Tsukada**, "Gwtc-2: Compact binary coalescences observed by ligo and virgo during the first half of the third observing run", *Physical Review X*, vol. 11, no. 2, Jun. 2021.

[39] B. P. Abbott, *et al.* including **L. Tsukada**, "Observation of gravitational waves from two neutron star–black hole coalescences", *The Astrophysical Journal Letters*, vol. 915, no. 1, p. L5, Jun. 2021.